D Hexadecimal Numbering

- Numbering Systems
- Relation Between Binary and Hex

While working with computers we are often required to use hexadecimal numbers. The reason for this is—everything a computer does is based on binary numbers, and hexadecimal notation is a convenient way of expressing binary numbers. Before justifying this statement let us first discuss what numbering systems are, why computers use binary numbering system, how binary and hexadecimal numbering systems are related and how to use hexadecimal numbering system in everyday life.

Numbering Systems

When we talk about different numbering systems we are really talking about the base of the numbering system. For example, binary numbering system has base 2 and hexadecimal numbering system has base 16, just the way decimal numbering system has base 10. What in fact is the 'base' of the numbering system? Base represents number of digits you can use before you run out of digits. For example, in decimal numbering system, when we have used digits from 0 to 9, we run out of digits. That's the time we put a 1 in the column to the left - the ten's column - and start again in the one's column with 0, as shown below:

```
0
1
2
3
4
5
6
7
8
9    last available digit
10   start using a new column
11
12
13
```

14
...
...

Since decimal numbering system is a base 10 numbering system any number in it is constructed using some combination of digits 0 to 9. This seems perfectly natural. However, the choice of 10 as a base is quite arbitrary, having its origin possibly in the fact that man has 10 fingers. It is very easy to use other bases as well. For example, if we wanted to use base 8 or octal numbering system, which uses only eight digits (0 to 7), here's how the counting would look like:

```
0
1
2
3
4
5
6
7     last available digit
10    start using a new column
11
12
 ...
 ...
```

Similarly, a hexadecimal numbering system has a base 16. In hex notation, the ten digits 0 through 9 are used to represent the values zero through nine, and the remaining six values, ten through fifteen, are represented by symbols A to F. The hex digits A to F are usually written in capitals, but lowercase letters are also perfectly acceptable. Here is how the counting in hex would look like:

0
1

```
2
3
4
5
6
7
8
9
A
B
C
D
E
F    last available digit
10   start using a new column
11
 ...
 ...
```

Many other numbering systems can also be imagined. For example, we use a base 60 numbering system, for measuring minutes and seconds. From the base 12 system we retain our 12 hour system for time, the number of inches in a foot and so on. The moral is that any base can be used in a numbering system, although some bases are convenient than others.

The hex numbers are built out of hex digits in much the same way the decimal numbers are built out of decimal digits. For example, when we write the decimal number 342, we mean,

```
   3 times 100 (square of 10)
+  4 times 10
+  2 times 1
```

Similarly, if we use number 342 as a hex number, we mean,

```
   3 times 256 (square of 16)
```